

Session 6.4

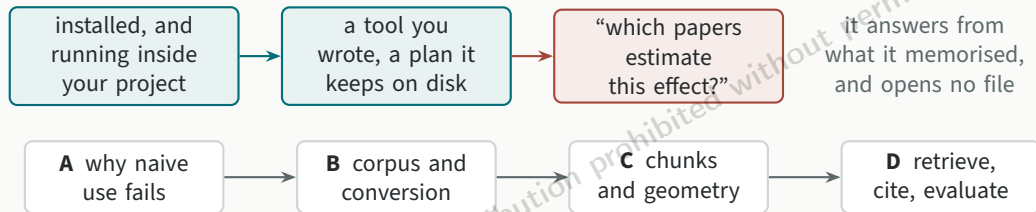
Build Your Own RAG: Give Your Agent a Library

Part four of five: why an agent cannot read your literature, what naive use costs, and a six-step recipe from a folder of PDFs to a cited answer — tied to Session 5 at each step



Zhigang Feng

Your Agent Runs Your Code, and Cannot Read Your Literature



- ▶ Decks 6.2 and 6.3 gave the agent hands: it runs commands, calls tools you wrote, and keeps a plan on disk.
- ▶ None of that put your literature in front of it. The shelf of papers on your own disk is still outside the loop.
- ▶ **Decks 6.2–6.3 closed the gap between advice and action. This deck closes the gap between action and evidence.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §1 and §2.1.

Session 6.4

Why Naive Use Fails

Three symptoms, two causes, the bill for pasting a paper, and three ways out



Three Symptoms of One Question

author, year and
title: all plausible

no such
paper

an argument, correct
somewhere else

a premise
is missing

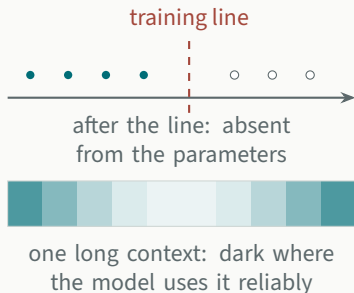
a fluent,
confident paragraph

no page,
no link

- ▶ Ask an unaugmented model which recent papers estimate your effect: three answers are possible.
- ▶ **Fabricated citation**: real scholars, a plausible year, a title in the right house style — and no such paper.
- ▶ **Misapplied method**: an argument right elsewhere, void here, and no warning given.
- ▶ **Confident and unsourced** is the third, and for academic writing it is a veto on its own.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.1.

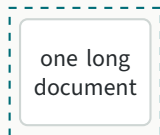
Two Causes: One “Does Not Have”, One “Cannot Use”



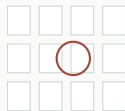
- ▶ **Knowledge cutoff** is a *does not have*: nothing past the training line is in the parameters.
- ▶ **Lost in the middle** is a *cannot use*: accuracy is highest at both ends of a long context and lowest in the middle.
- ▶ Two biases stack — recency, from causal attention, and primacy, from where instructions sit in the training data.
- ▶ **The constraint is effective attention, not capacity.**

Source: Liu et al. (2024), “Lost in the Middle”, *TACL* 12, 157–173.

Which Problems a Wider Window Actually Solves



capacity
binds: a wider
window helps

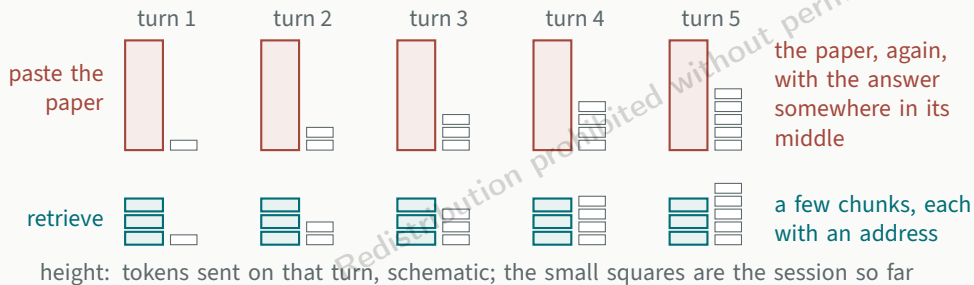


location and
provenance
bind: width
buys nothing

- ▶ Sort the information need into two kinds before choosing the tool.
- ▶ **Read one long document:** a working paper, a long policy file. Capacity binds, and a wider window genuinely helps.
- ▶ **Locate and cite:** a few passages somewhere in the whole corpus. No width helps; you need an address and a source.
- ▶ **If the task is both, split it** — and note that a wide context is paid for again at every question.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.1.

The Cost of Naive Use: Paste the Paper, Pay Every Turn



- ▶ Paste a paper once and every later question re-sends it, on any topic (Deck 6.2).
- ▶ The passage you need sits in the middle, where it is used least reliably (Deck 5.3; f3).
- ▶ **Retrieval sends only what the question needs, with each chunk's address: the bill stops growing with the paper.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.1–§2.2; Liu et al. (2024), *TACL* 12.

Three Ways to Give a Model Your Literature

paste it

the paper in the prompt,
re-sent every turn

context: this turn and
every turn after (f5)

index it

cut, embed, re-
trieve; cite the chunk

retrieval: an address
per passage, built once

send an agent

it searches the web and
writes a cited review

deep research: the
misses look like hits

- ▶ Rao et al. (2026), ten models and agents: 3–13 % of citation URLs are hallucinated, 5–18 % non-resolving. Onweller et al. (2026), fourteen models: link validity above 94 % and relevance above 80 %, yet only 39–77 % factual accuracy.
- ▶ **The misses look like hits:** a live link, a relevant page, a wrong sentence. Your own index cites or declines — this deck.
- ▶ Not on this list: fine-tuning changes how the model speaks, not what it can cite (f7).

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.2; Rao, Wong & Callison-Burch (2026), arXiv:2604.03173; Onweller et al. (2026), arXiv:2605.06635; both preprints, read 23 September 2026.

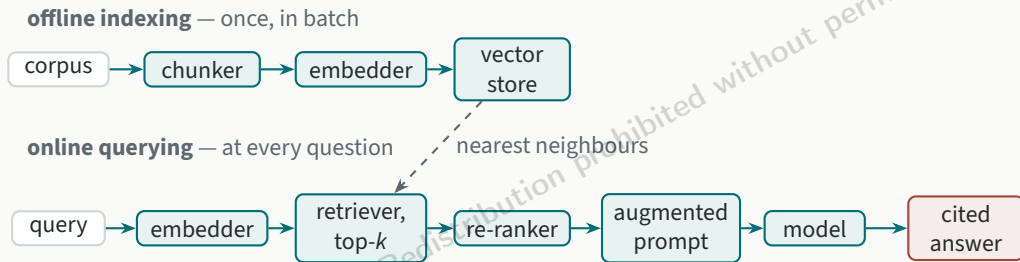
Fine-Tuning Versus Retrieval: Six Dimensions

dimension	fine-tuning	retrieval
compute	a training run, on GPUs	index once, then query
update delay	days: retrain and redeploy	minutes: add a document
attribution	none, by construction	built in: one tag per claim
style, format	strong: voice, terms, layout	weak: the base model writes
hallucination	limited: recall is parametric	stronger: held in the context
typical use	imitation, house terminology	live corpora, policy questions

- ▶ The table points one way for literature questions; the hybrid fine-tunes the embedder, not the writer.
- ▶ **Fine-tuning’s comparative advantage is style, not knowledge.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.2, the six-dimensional trade-off.

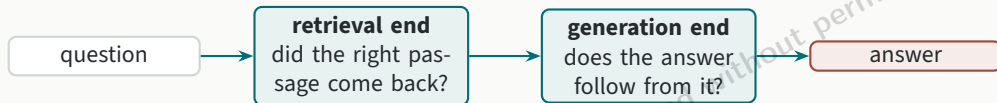
The Pipeline in One Picture



- ▶ **Offline indexing** runs once: chunk, embed, store the vectors and their metadata.
- ▶ **Online querying** runs per question: embed, retrieve, re-rank, inject, answer.
- ▶ **Index once, query many**: a fixed cost against a marginal cost near zero; each question ships 5 to 20 chunks, not the corpus.
- ▶ **Both chains use one embedder**: similarity exists only inside one space.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3; Lewis et al. (2020), *NeurIPS* 33, 9459–9474.

Score It Before You Build It: Two Ends, a Score for Each



a good score at one end and a bad one at the other is not a good system — it is a system that fails in a way you have not measured

- ▶ An economist has met this shape before, and it runs the other way. A doubly robust estimator is safe if *either* the outcome model or the propensity model is right; here a good score at one end buys nothing if the other fails.
- ▶ Write the score first, with questions the corpus cannot answer: RAG's V0 gate (Deck 6.3).
- ▶ **Both ends must pass. They fail for different reasons and are repaired in different places.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.4.

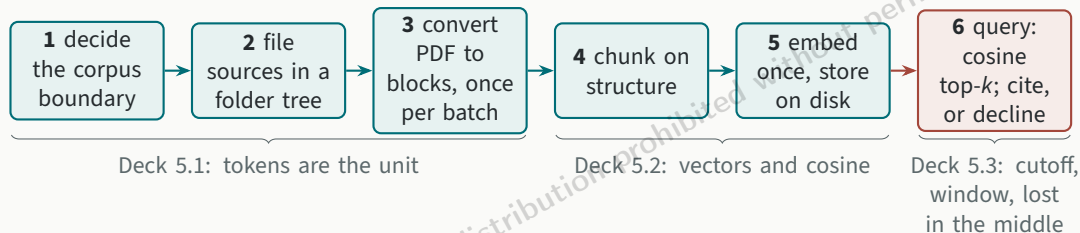
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Stages 1–2: Corpus and Conversion

Deck 5.1 made tokens the unit: the model reads text, so the first jobs are deciding which text, and turning files into it



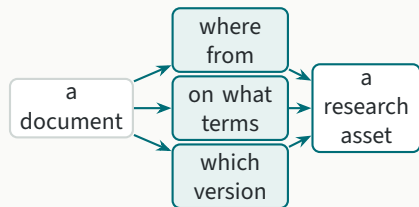
The Recipe, in Six Steps



- **Steps 2 to 6 are engineering you can copy. Step 1 is research only you can do.**
- f8 drew the pipeline on paper; this part walks one in daily use — 162 documents, 74,557 chunks, one index on disk — and rung two is the control condition Deck 6.5 has to beat.
- **The goal is not to admire it. It is to leave with the recipe for your own corpus.**

Source: Instructor's working retrieval infrastructure, read from source on 23 September 2026; Miao & Feng, *AI for Economists*, ch. 15, §2.5.

The Corpus Is a Research-Design Choice

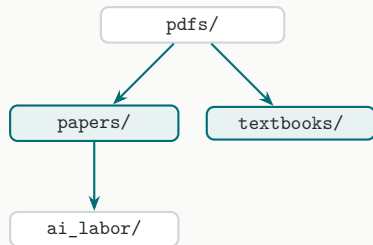


any one missing: a pile
of files, not a corpus

- ▶ The corpus is the system's whole knowledge: what is left out cannot be retrieved.
- ▶ Ready-made, one per field: a central bank's archive, a filings database, a working-paper archive.
- ▶ **A corpus with no licence is not a research asset:** no provenance, no terms, no frozen version, nothing a referee can re-run.
- ▶ This session: 245 public-domain statements, 1994-02-04 to 2026-06-17. The worked system: 162 documents, 88 of them papers.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3 and §2.5; this session's frozen corpus manifest; the working system read from source on 23 September 2026.

The Folder Tree Is the Metadata; the Hash Prevents Repeat Work



a document's path *is* its topic

- ▶ There is no separate topic field to maintain. Filing a paper in a folder is how it gets classified.
- ▶ A manifest row carries a content hash, so a rerun passes over converted sources; each numbered stage reruns alone.
- ▶ **The cheapest metadata schema is the one you cannot forget to update.**

Source: Instructor's working retrieval infrastructure, read from source on 23 September 2026.

Stage 2: A PDF Becomes Readable Context

```
export LC_ALL=en_US.UTF-8                # CJK file names, spaces in paths
STAGE="$(mktemp -d)"; for f in "$IN"/*.pdf; do n="$(basename "$f" .pdf)"
[ -d "$OUT/$n" ] || ln -s "$(readlink -f "$f")" "$STAGE/"; done # unconverted
mineru -p "$STAGE" -o "$OUT" -b vlm-auto-engine # the converter runs ONCE per batch
ls "$OUT" | wc -l; wc -w "$OUT"/*/vlm/*.md # files kept? words kept?
```

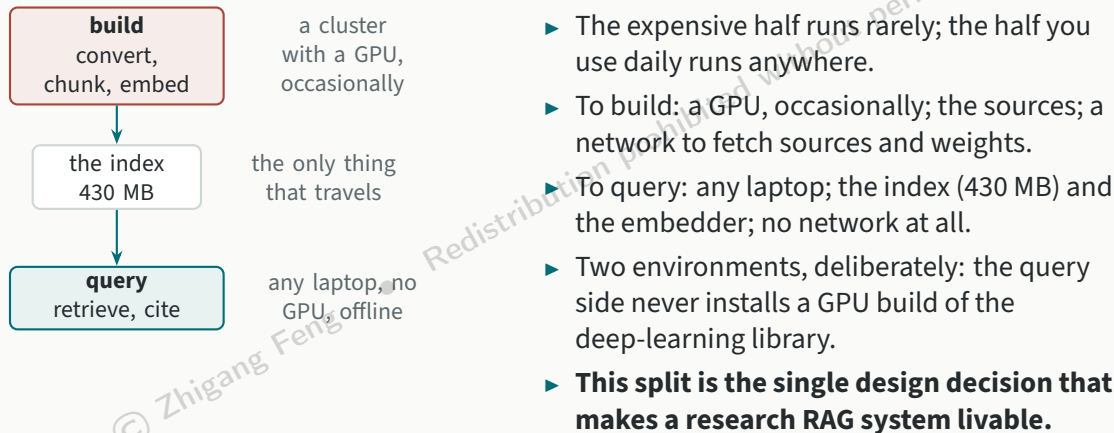
a 54-page paper, one A100 GPU

engine start-up, model load	110 s , <i>fixed</i>
inference, all 54 pages	0.8 s <i>a page</i> , <i>marginal</i>

- Conversion has **a fixed and a marginal cost**: called once *per file*, thirty PDFs cost fifty-five minutes of model loading around twenty minutes of work. Stage the batch, call it once.
- Normalise early: every format becomes one block record. The silent failure in the logs: a concurrency default deadlocks the engine.

Source: Course conversion benchmark, recorded July 2026 (the benchmark folder no longer exists; the three numbers are carried from the legacy frame of that date).

The Design Rule: Build on a GPU, Query Anywhere



Source: Instructor's working retrieval infrastructure, read from source on 23 September 2026.

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Stages 3–4: Chunks and Geometry

Deck 5.2's vectors and cosine: cut the text into citable units, then place each one where nearness means meaning



Why Chunk at All; Five Strategies, Each More Respectful of Structure

“the effect is significantly negative”

which effect?
which sample?

one block,
three topics

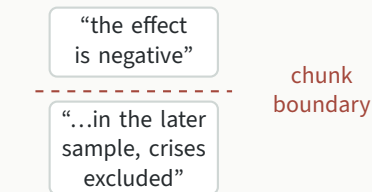
precision
diluted,
the prompt
crowded

the unit of retrieval is
the unit of evidence

- ▶ **Chunking** cuts the corpus into the units retrieval returns. Too small, no context; too large, several subjects and a crowded prompt.
- ▶ **Fixed length, sliding window, recursive character**: each breaks at a more natural boundary.
- ▶ **Semantic**: cut where similarity between neighbours drops.
- ▶ **Document-aware**: headings, tables, metadata; on economic text, almost always worth its cost.
- ▶ **Cut at the unit you would be willing to cite.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3.

The Economist's Trap: A Claim Cut From Its Condition



- ▶ Economic text is already structured: a statement has an action, an assessment and a vote.
- ▶ The trap is cutting a claim away from the condition it holds under: the sample, the specification, the date.
- ▶ Document-aware chunking keeps the unit whole and puts the heading in the metadata.
- ▶ **A chunk that drops its conditions is worse than none:** it reads as general and is not.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3, on structured economic corpora.

Headings Are Hard Boundaries, and What One Chunk Carries

3.2 Identification



...block, block, block

...up to 1200 characters



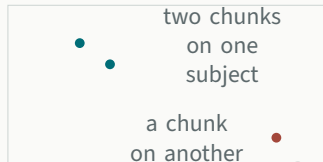
3.3 Estimation

a new chunk
starts here

- ▶ Blocks accumulate to about 1200 characters, with a 150-character overlap and a 200-character floor.
- ▶ The overlap is whole blocks walked backwards: it never cuts a sentence, and *never* crosses a heading.
- ▶ Each chunk carries a **locator**, a page range or a section path: that is why an answer can cite.
- ▶ **A chunk that spans two sections is a chunk that answers neither question.**

Source: Instructor's working retrieval infrastructure, read from source on 23 September 2026.

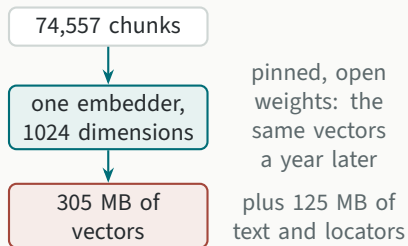
From Chunks to Geometry



- ▶ **Document embedding:** one vector for a whole chunk, $\mathcal{E}(x) \in \mathbb{R}^{d_e}$, with d_e usually between 384 and 4096.
- ▶ Deck 5.2 drew this geometry: nearness stands for closeness in meaning.
- ▶ So do its warnings: two filings from different industries look alike when the boilerplate does.
- ▶ **You do not implement this:** call an encoder, feed the chunks in batches, keep the vectors.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3; Deck 5.2 of this course.

Choosing an Embedder; One Bilingual Model, Vectors on Disk



- ▶ Choose on four things: the dimension you can store, the corpus's languages, domain fit, and whether the weights are yours to pin.
- ▶ One model embeds corpus and query alike; change the encoder and the index is stale, not slightly wrong.
- ▶ Bilingual corpus — 158 English documents, 4 Chinese — so the embedder is too.
- ▶ **Open weights are a replication requirement:** a hosted endpoint can be upgraded under you; a pinned version can be re-run. The build refuses vectors from another model.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3; the working system read from source on 23 September 2026.

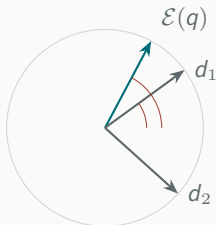
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Stage 5: Retrieve, Cite, Evaluate

Deck 5.3's three limits — the cutoff, the window, lost in the middle — are what the last step is built around



Similarity Search, and What a Vector Store Is For



cosine compares angles,
so chunk length drops out

- ▶ Compare query and chunk by the angle between their vectors:

$$\text{sim}(q, d_i) = \frac{\mathcal{E}(q)^\top \mathcal{E}(d_i)}{\|\mathcal{E}(q)\| \|\mathcal{E}(d_i)\|}.$$

- ▶ **Top-*k***: the retriever returns the 5 to 20 nearest chunks; the worked system returns 8. A **vector store** holds the vectors and the metadata and answers that query fast.
- ▶ **Nothing here understands the question**; it measures angles.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3; the working system read from source on 23 September 2026.

Two Lists, One Ranking: Keywords and Rank Fusion

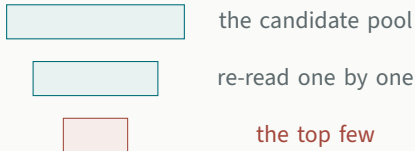
dense	sparse	fused
A	D	A
B	A	D
C	E	B

A tops one list and is second in the other, so it wins the merge

- ▶ Dense retrieval can walk past an exact string; **BM25** counts the literal term. Two lists, two kinds of score, in different units.
- ▶ So merge by *rank*:
$$\text{RRF}(d) = \sum_{r \in R} 1/(\kappa + r(d)), \text{ with } \kappa = 60.$$
- ▶ The paper, and the lab's code, call it k ; κ here keeps it apart from top- k .
- ▶ Agreement across lists beats one high rank; low-ranked chunks still count.
- ▶ **Fixed in a pilot and never re-tuned:** a default, not a finding.

Source: Cormack, Clarke & Büttcher, *Proceedings of SIGIR '09* (2009), 758–759.

Re-Ranking: A Cheap Screen, Then an Expensive Read



editors screen, referees
read: the expensive step
never runs on everything

- ▶ The first pass is fast and approximate: one stored vector per chunk against one vector for the query.
- ▶ **Re-ranking** hands the shortlist to a slower model that reads query and chunk together, then keeps the best few.
- ▶ **The screening step decides what the careful step can ever see:** a chunk missed in the first pass is not recoverable later.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3, on coarse screening and fine ranking.

Stage 5: Retrieve, Then Cite — the Augmented Prompt Has Four Parts

system: answer from the context only; tag every claim; if it is not there, say so

context: the top 8 chunks, each with identifier, title, page range and score

question

format: how to cite, how long

- ▶ The query is embedded by the same model, scored by cosine against every vector; the top 8 come back. A topic filter applies *after* ranking, so a narrow literature never starves the result.
- ▶ **Grounding** is the system line: answer from the context, fill no gap from memory; the template turns evidence into an answer.
- ▶ The only place a generative model is used. **Everything upstream exists to put good evidence in front of it.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3; the working system read from source on 23 September 2026.

The Citation and the Refusal, and a Worked Answer

Q. What did the Committee say at that meeting, and what did the ten-year yield do next?

A. one clause [doc-07]; a second clause [doc-12].

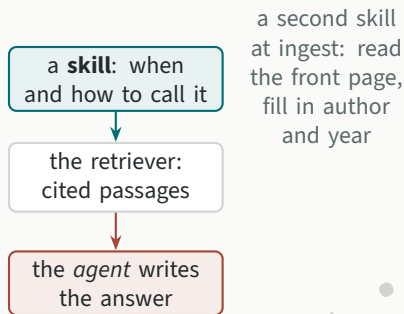
Q. And fiscal policy? **A.** Not in the retrieved context.

each clause carries the identifier of the chunk it came from, assigned at chunking time

- ▶ One stable identifier per chunk: required by the template, resolved back to the page.
- ▶ In the lab the refusal is a number: the best evidence score among the top 5 chunks against a threshold of 0.218; 4 of 6 unanswerable questions are turned away.
- ▶ **“Not in the context” is the system working:** a refusal costs one question, a fabrication costs the paper.
- ▶ **A tag proves provenance, not correctness:** whether the right chunk came back is measured, not assumed — the last part of this deck.

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3 and §2.5; the session’s lab protocol.

Making It a Tool Your Agent Can Reach



a second skill
at ingest: read
the front page,
fill in author
and year

- ▶ Deck 6.2 defined a skill as a folder the model opens when needed. This is one, pointed at a real index.
- ▶ The retriever never calls a language model: it prints evidence and an instruction — answer only from this, cite every claim, say so if the evidence is weak.
- ▶ **The generator is the agent you already have: the library is one more tool.**
- ▶ Four seams, cut in advance: a format is one converter, a chunker one strategy, a store one interface, settings one file.

Source: Instructor's working retrieval infrastructure, read from source on 23 September 2026.

Faithfulness: The One That Matters Most

- ▶ Break the answer into claims. For each, ask whether the retrieved passages actually support it.
- ▶ A fluent answer that adds one unsupported sentence is more dangerous than an answer that refuses.
- ▶ **This is the score that stands between a citation and a fabricated one.**
- ▶ In the lab you mark one answer by hand:
 $F = \#\{j : C \models c_j\} / m$, the share of its atomic claims that the context supports.

claim 1: supported

claim 2: supported

claim 3: nowhere
in the passages

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.4; the course lab's gold set.

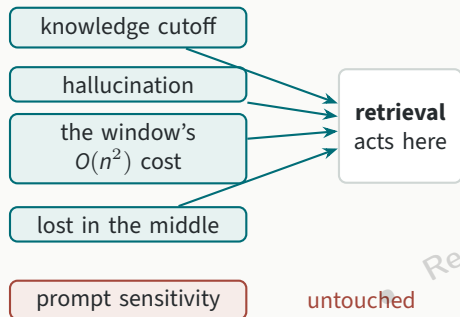
Five Ways It Breaks, and Where Each Sits

What you see	Where it sits	What repairs it
nothing relevant comes back	the index	the corpus boundary
the right passage ranks fourth	the ranking	a second scorer, or re-ranking
a claim lost the condition it held under	the chunker	boundaries that respect structure
a confident answer about a year you never indexed	the query	a refusal rule
a confident answer from the wrong meeting	an ambiguous query	ask it back before answering

The lab produces three on purpose: a stale index, a cut inside the decision sentence, an ambiguous question answered from the wrong meeting. **The symptoms read alike from outside. Diagnose by asking which stage could have produced this one.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.4.

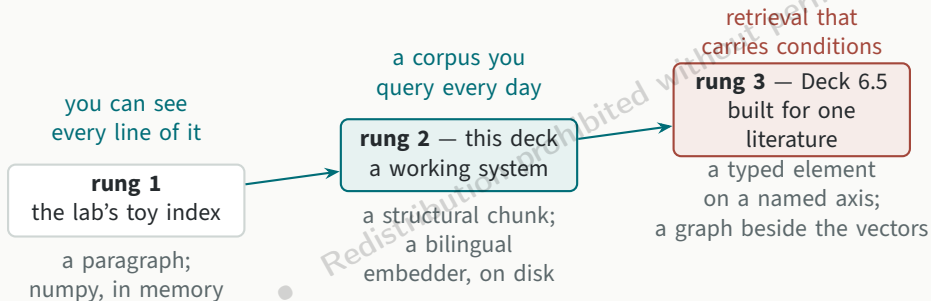
The Five Limits Session 5 Left Standing, Now Four



- ▶ Deck 5.3 closed on five limits of a pretrained model, all architectural: a larger model does not remove them.
- ▶ The six steps repair four by one move: the knowledge left the parameters for an index you can rebuild today.
- ▶ **The fifth is untouched.** Prompt sensitivity belongs to the prompt, not the evidence.
- ▶ Two limits remain: the recipe refuses by a sentence in its prompt (the lab, by a threshold), and a dense vector cannot tell you when it is wrong.

Source: Miao & Feng, *AI for Economists*, ch. 13, §6.1, and ch. 15, §2.5.

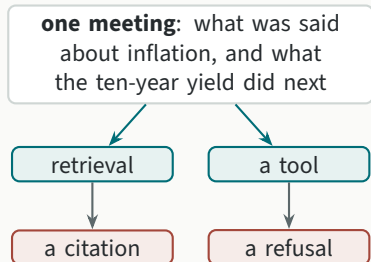
Three Rungs: Toy, Working, Research



- ▶ This deck built rung two on a corpus of your own; Deck 6.5 is GRAM, built for one literature.
- ▶ **Each rung answers what the rung below answers badly, and costs more to keep standing. Climb when the rung below is boring.**

Source: Miao & Feng, *AI for Economists*, ch. 15, §2.3 and §2.5.

Your Turn: Lab Blocks C, D and E



drop any one and the task is still answerable — fluently, and wrongly

- ▶ **C**, build the index: 245 statements, chunks that name their statement, two lists fused by rank.
- ▶ **D**, answer or refuse: a threshold fitted on the graded set, and three failures produced on purpose.
- ▶ **E**, faithfulness by hand: split one answer into claims and mark each one supported or not.
- ▶ **Self-paced, after class, offline: a CPU, no account, fixed seeds.**

Source: The session's lab, blocks C to E; Session 2's frozen yield panel.